

Data Wrangling 🤠

Expressions

Lesson 3

Expressions

- The building blocks of a program
- When you **run** code, expressions will evaluate to a typed value

Demo: Literals

```
>>> 99 # int literal
99
>>> 100.1 # float literal
100.1
>>> "cowboy" # str literal
'cowboy'
```

Demo: Operator Expressions

```
>>> 1 + 2 # operator expressions
```

```
3
```

```
>>> 1.5 + 1.5 - 1
```

```
2.0
```

```
>>> 3 / 2
```

```
1.5
```

```
>>> 3 * 2
```

```
6
```

Numerical Operators

Symbol	Name	Example
**	Exponentiation	<code>2 ** 3</code> → <code>8</code>
*	Multiplication	<code>10 * 2</code> → <code>20</code>
/	Division	<code>8 / 3</code> → <code>1.6</code>
//	Integer Division	<code>8 // 5</code> → <code>1</code>
%	Remainder	<code>8 % 5</code> → <code>3</code>
+	Addition	<code>1 + 1</code> → <code>2</code>
-	Subtraction / Negation	<code>3 - 1</code> → <code>2</code>

Demo: String Concatenation

```
>>> "hey there" + "cowboy" # concatenation
```

```
'hey therecowboy'
```

```
>>> "I was born in " + 2010
```

```
Traceback (most recent call last):
```

```
  File "<python-input-30>", line 1, in <module>
```

```
    "I was born in " + 2010
```

```
~~~~~^~~~~~
```

```
TypeError: can only concatenate str (not "int") to str
```

```
>>> "201" + "1"
```

```
'2011'
```

Demo: Boolean Expressions

```
>>> 5 > 3
```

```
True
```

```
>>> 3 <= 3
```

```
True
```

```
>>> 5 <= 3
```

```
False
```

```
>>> 5 == 5
```

```
True
```

```
>>> 3 != 5
```

```
True
```

```
>>> 5 != 5
```

```
False
```

Demo: String Comparison

```
>>> "a" < "b"  
True  
>>> "ant" < "cat"  
True  
>>> "cat" != "dog"  
True
```

Relational Operators

Symbol	Meaning	True	False
<code>==</code>	equal	<code>1 == 1</code>	<code>1 == 2</code>
<code>!=</code>	not equal	<code>1 != 2</code>	<code>1 != 1</code>
<code>></code>	greater than	<code>2 > 1</code>	<code>1 > 2</code>
<code>>=</code>	at least	<code>2 >= 1</code>	<code>0 >= 1</code>
<code><</code>	less than	<code>1 < 2</code>	<code>2 < 1</code>
<code><=</code>	at most	<code>1 <= 2</code>	<code>2 <= 1</code>

⚠ *The equal-to operator is `==` not `=`*